

A Set-Transformer Agent for the Pokémon TCG: Architecture, Evidence, and What Is Not Yet Justified

Research lane

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Abstract

We describe the architecture of a set-transformer policy for the Pokémon Trading Card Game competition, and — deliberately given equal weight — the boundary between what its design choices are known to buy and what is currently asserted without a measurement behind it. Three changes distinguish the current architecture (v2) from its predecessor: a family-conditioned (FiLM) numeric projection, multi-seed pooled attention with per-consumer read-outs, and per-column input standardisation folded so that a warm start is exact. Each has a stated mechanism; we check the mechanisms against the literature and against the code, and we report one case where the project’s own design document rejects a component on evidence that does not support the rejection, and another where two components are *structurally inseparable* in the implementation and therefore cannot be ablated as planned. We close with the measured between-seed variation, which is the single most transferable result we have: architecture differences of the size currently being reported sit close enough to seed noise that single-run comparisons should not be published as findings.

1 Evidence tags

Every quantitative claim in this document carries one of five tags. A claim without a traceable measurement does not appear.

[THEOREM]	follows by algebra or from a cited theorem
[PRACTICE]	established practice in the literature, not a theorem
[MEASURED]	measured in this project, with the producing run named
[SPECULATIVE]	plausible mechanism, no measurement
[UNSOURCED]	a number in circulation whose source could not be located

2 The token surface

The game state is presented to the network as a *set* of typed entity tokens, not a sequence. State tokens and one token per legal option occupy a single sequence, so all options are scored in one forward pass. Legality is a property of the input: an illegal action is never constructed and therefore has no logit.

Padding is masked, and this is not a tuning choice. [theorem] [measured] Unmasked attention places every PAD slot in the softmax denominator, so the representation of every *real* token becomes a function of the padding width — a property of the batch, not of the game state. Training collates to a state width of 160; the serving featuriser pads to 192. We held real token content byte-identical and varied only the number of trailing PAD slots, using trained weights in evaluation mode:

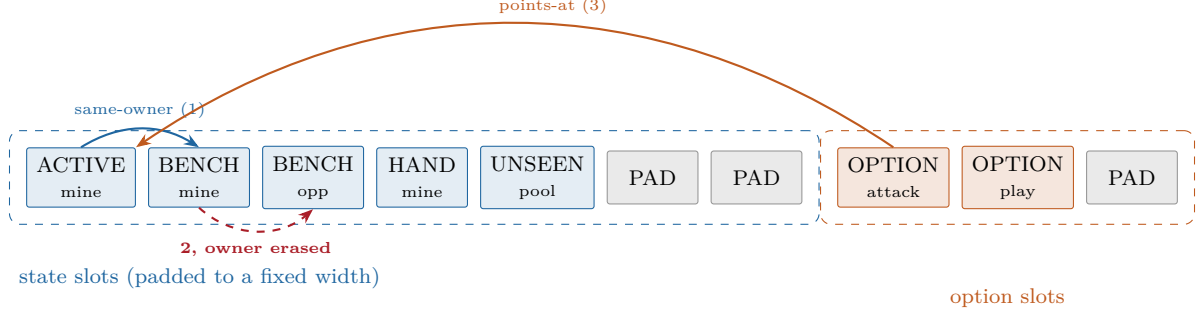


Figure 1: Token layout and typed relations. Relations are attribute-derived (same-owner, same-zone, active-vs-active) or instance-specific (the `option_ptr` edge naming which state token an option refers to). The dashed red edge is a defect, not a design: see §6.

	argmax changed, train $\rightarrow$ serve	mean max $ \Delta \text{logit} $
masked	0/300 = 0.00%	1.71×10^{-6}
unmasked	37/300 = 12.33%	1.355

The masked residual is floating-point kernel-tiling noise. The unmasked figure means an unmasked model *serves a different function than the one it trained*. This also settles a separate claim: bucketing the pad width to $\{64, 96, 160, 232\}$ for a large serving speed-up is representationally free *under masking* and is not free without it.

3 The numeric projection, and where FiLM enters

3.1 The defect being repaired

Numeric columns are laid out per family, so the same column index carries different quantities on different token families: column 8 is `bench_slot_0` on a BENCH token and hand position 0 on a HAND token. Permutation attribution of that single column, over 600 held-out frames, is 0.0241 from HAND against 0.0003 from BENCH and 0.0004 from ACTIVE. [MEASURED]

In v1 the family enters the token representation *additively*, after a single shared linear map applied to all families.

An additive family embedding cannot modulate a linear map. [theorem] Writing z for the standardised numeric vector, f for the family and $e(\cdot)$ for the additive embeddings,

$$x_{v1} = e_{\text{fam}}(f) + e_{\text{own}}(o) + e_{\text{card}}(c) + Wz + b, \quad \frac{\partial x_{v1}}{\partial z} = W,$$

which is independent of f . The family term shifts the output; it cannot change how the map reads its input. The claim as stated in the project’s design notes is therefore correct.

3.2 What FiLM does, exactly

FiLM [1] conditions a network on side information by a feature-wise affine transform. In this implementation it is applied to the *input* of the projection, with per-family gain and offset vectors of the same width as the numeric surface:

$$x_{v2} = \dots + W(\gamma_f \odot z + \beta_f) + b, \quad \frac{\partial x_{v2}}{\partial z} = W \text{diag}(\gamma_f).$$

So the contribution of column j under family f is $W_{:,j} \gamma_f[j] z_j$.

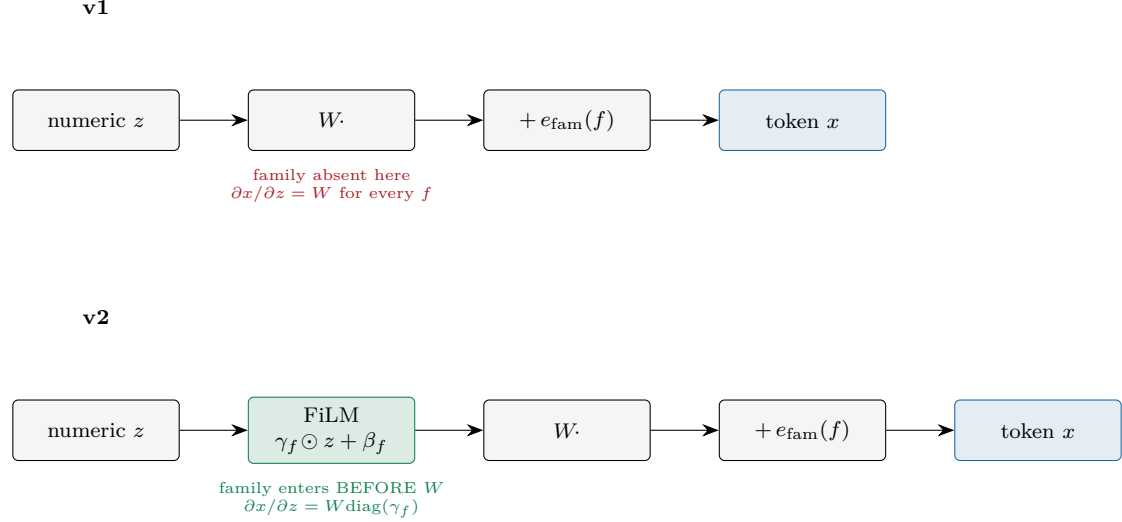


Figure 2: Where family conditioning enters. In v1 the family is added after the projection and cannot alter it. In v2 FiLM multiplies the projection’s input, so each (family, column) pair gets an independent scalar gain.

The precise capability, and the precise limit. [theorem] FiLM gives an *independent scalar gain per (family, column)*. It can therefore switch column 8 on for HAND and off for BENCH, eliminating interference between two families that share a column index. It cannot give column j a different *direction* in $\mathbb{R}^d$ per family: that direction is $W_{:,j}$ for every family, and FiLM moves the contribution only along the line $\text{span}\{W_{:,j}\}$ (a negative gain reverses it). Formally, FiLM is exactly the subfamily of per-family projections constrained to $W_f = W \text{diag}(\gamma_f)$.

Is FiLM the right instrument here? Yes, on the evidence that exists. The project’s ground-up design document rejects FiLM, arguing that column 8 on HAND and on BENCH are “unrelated quantities needing different directions.” The measurement offered in support (0.0241 versus 0.0003) does not show that. An attribution of 0.0003 is a column the family does not use; that is an *interference* pattern, which a gate repairs, not a *direction conflict*, which a gate cannot. The rejection is stated more strongly than its cited evidence supports. The cost argument favours FiLM sharply: $2 \cdot N_{\text{fam}} \cdot w = 12,768$ parameters against $N_{\text{fam}} \cdot d \cdot w \approx 2.45\text{M}$ for per-family matrices, a factor of roughly 192. [MEASURED]

The experiment that would decide it, and it has not been run. [speculative] Train one arm with unconstrained per-family projections W_f and measure the pairwise cosine similarity of $W_f[:, j]$ across families, restricted to the high-attribution columns. If those directions are near-parallel, FiLM is sufficient and the additional parameters are waste; if they are not, FiLM is leaving representable structure on the table. Nobody has measured this.

4 Pooling: one seed, or one per consumer

Set Transformer [2] pools a set by attention against k learned seed vectors, $\text{PMA}_k(Z) = \text{MAB}(S, \text{rFF}(Z))$ with $S \in \mathbb{R}^{k \times d}$. Multi-seed pooling is the original formulation, not an extension of it: the paper uses $k > 1$ wherever k outputs are wanted. [PRACTICE]

Assigning seeds to consumers is a routing choice, and it is the part without a citation. [speculative] That k seeds exist is standard; that seed 0 feeds the policy, seed 1 the value head and seed 2 the auxiliary bank is a design decision specific to this project.

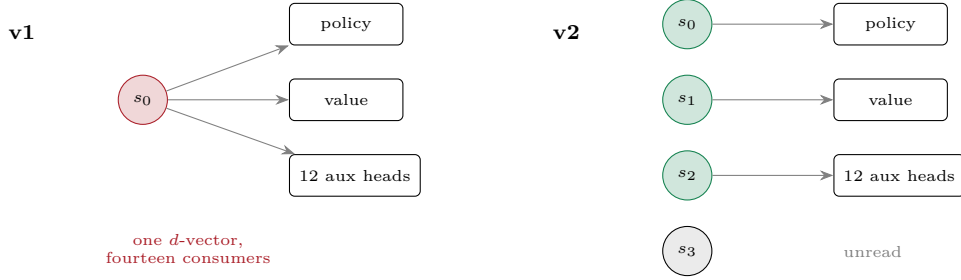


Figure 3: Pooling fan-out. v1 routes a single pooled vector to every consumer. v2 gives the policy, the value head and the auxiliary bank their own seed, so each consumer’s gradient shapes its own read of the board.

It is well-motivated — disjoint read-outs mean each seed receives gradient from exactly one consumer, which forces the specialisation that otherwise merely has permission to emerge — but no published result is being relied on, and none should be cited as if it were.

k seeds give k rank-one reads, not a full-rank read. [theorem] A single-layer PMA is one softmax-weighted average, i.e. a rank-one summary of the token set; k seeds give k of them. This is a partial answer to the objection that a pooled read cannot express conjunctions over the board (“their active is a two-prize threat *and* I have exactly enough energy *and* they have one prize left”), not a complete one.

4.1 A structural confound: FiLM and multi-seed pooling cannot be separated in the current code

This is the most consequential implementation finding in this document.

- The constructor **raises** if `arch_v2` is set with fewer than three pool seeds, because the v2 forward reads seeds 0, 1 and 2 by index.
- The pooling module is built as $\text{PMA}(d, h, \text{seeds} = k)$ *only* when `arch_v2` is on; under v1 the seed count is ignored and pooling falls back to a single seed.

Therefore `arch_v2` $\Leftrightarrow$ multi-seed pooling, enforced at construction. There is no configuration that isolates FiLM from multi-seed pooling, in either direction. A side effect worth its own note: the v1 arm’s manifest records `pool_seeds: 4` while the code ignored the flag and built a single seed — a manifest that documents an intent the run did not execute, which is the same class of defect as a device flag recorded by an implementation that cannot use it.

What this does and does not invalidate. It makes the v1-versus-v2 comparison *undecomposable*: FiLM, pooling seeds and standardisation moved together and no configuration separates the first two, so no per-component attribution can be drawn from it. It does *not* invalidate single-variable runs that toggle one flag against a *fixed* v2 baseline — relations on or off, head geometry, model width. Those remain attributable to their own variable; they were never capable of decomposing the v2 package, and should not be described as if they were. Decoupling the flags is a change to a file that is frozen while runs import it, so it belongs in the next generation, together with the per-family W_f cosine test of §3.2.

5 Per-column standardisation

Raw numeric columns span a measured $39,570\times$ in per-column standard deviation; a signed $\log(1 + |x|)$ leaves $891\times$; the learned projection’s per-column norms span only $9.27\times$, so the

network does *not* compensate for the residual imbalance. Five of 160 columns carry half the input variance entering the projection, while 52 binary flags — most of the hand-engineered feature surface — carry 2.3%. [MEASURED]

The fold is exact. [theorem] Standardising $x' = (x - \mu)/\sigma$ and absorbing it into an existing checkpoint by

$$W'_{:,j} = W_{:,j} \sigma_j, \quad b' = b + W\mu$$

gives $W'x' + b' = W(x - \mu) + b + W\mu = Wx + b$, so the warm start is bit-identical at initialisation while gradients are equalised from the first step. Two constraints are load-bearing: $\sigma_j = 0$ must pass through unchanged rather than divide, and σ must be robust (median/IQR with a floor), because spike distributions are common in this feature set — a column that is 0 on most rows and 40 on the rest is scaled *into* an outlier by a naive standard deviation rather than out of one.

6 The relation that erases what it is for

Attribute relations are assigned by two successive overwrites: same-owner first, then same-zone *unconditionally*. Because family and owner are separate tags, every pair sharing a family receives the same-zone identifier regardless of owner — so a bench token and the opposing bench token can carry the identifier that is supposed to distinguish them. The same logic appears in a second code path. Measured over 400 real rows, excluding padding: [MEASURED]

same-owner pairs relabelled to same-zone	371,200/1,200,682 = 30.92%
same-zone pairs that are in fact cross-owner	13,904/385,104 = 3.61%

The defect is real and should be repaired in both code paths. It is *not*, however, a sufficient explanation for the relational bias being inert. That bias is a scalar per (relation, head) — 768 parameters across all twelve blocks — and trains to an absolute maximum of 0.035 in a logit space of unit scale, flipping 0.2–0.3% of decisions. The instance pointer, which delivers a relation as a full d -dimensional input feature, uses 294,912 parameters and flips 36.2%. [MEASURED] The honest reading is about *delivery mechanism*: relations supplied as input features work here; relations supplied as scalar attention biases do not. A 3.61% mislabelling rate on one identifier does not account for the gap.

7 Seed variation, and why single-run comparisons should not be published

There are two seed bands in this project and they are not interchangeable. One is the spread of *offline top-1* across training seeds; the other is the spread of *arena win rate* across match seeds. They differ by roughly an order of magnitude, for the ordinary reason that a win rate over a few hundred paired matchups is far noisier than an accuracy over fifteen thousand rows. A band quoted without naming its quantity invites comparison against something nobody measured.

band	quantity	between-seed SD	source
offline	contested top-1	0.15–0.27 pp	20 training runs, Fig. 4
arena	win rate	2.51–2.52 pp	16 match seeds, Fig. 5

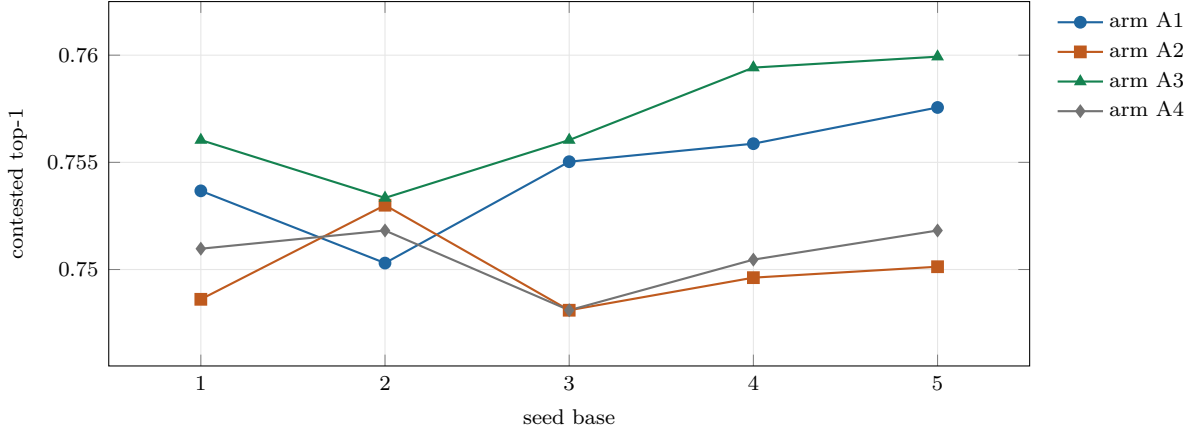


Figure 4: Twenty runs: four architecture arms at five seed bases each, scored on contested top-1 (forced single-option decisions excluded). Between-seed standard deviation is 0.15–0.27 pp and the full five-seed range is 0.37–0.73 pp. Arm A1 beats arm A2 on the five-seed mean (0.7545 against 0.7499, a 0.46 pp effect) and at four of the five seeds — but at seed base 2 the ordering reverses. A comparison run at a single seed therefore returns the wrong sign on a real effect one draw in five.

Applying this to the current v1-versus-v2 result. Both arms were trained on identical splits (4,920 train / 868 validation games; 19,900 validation rows of which 15,804 are contested), at the record shape, from the same seed, for two epochs:

	contested top-1	Brier	headline top-1
v1	0.7281	0.4144	0.7598
v2	0.7371	0.3975	0.7669
Δ	+0.90 pp	−0.0170	+0.71 pp

The margin is larger on contested decisions than on the headline, which is the right direction: v2 gains where a choice actually exists. **[MEASURED] This number is provisional pending the paired interval below, and should be cited that way.** Three qualifications belong beside it, and none of them is optional.

1. **$n = 1$ seed per arm.** The +0.90 pp exceeds the largest five-seed range we have measured (0.73 pp), which is encouraging, but that band comes from a different corpus, a different width and head count, and ten epochs rather than two. Early training is the noisier regime, so the band almost certainly understates seed noise here. A second seed is the minimum.
2. **The comparison is paired on data and split but not on initialisation,** since v2 adds parameters and the random stream diverges.
3. **The correct test has not been run.** Because both models are scored on the same games, the statistic that settles this is the per-game difference in contested accuracy, bootstrapped over the 868 validation games — not a row-level interval, which treats 15,804 correlated rows as independent and is optimistic by roughly 4× here. This costs one evaluation pass that writes per-game accuracies.

And the result is a package, not a component. The v2 arm moved FiLM, pooling seeds and standardisation together. Per §4.1, the first two cannot be separated at all in the current code. The defensible claim today is “the v2 package beats v1 by +0.90 pp contested at one seed”; no per-component attribution is available.

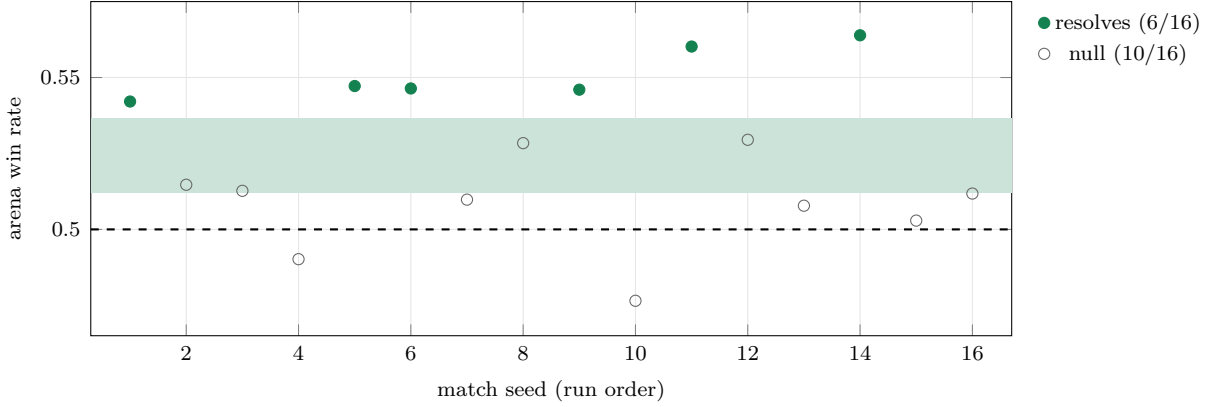


Figure 5: The same comparison run at sixteen match seeds, 256 paired matchups each. Dashed line is no effect; the shaded band is the pooled 95% interval, $[0.5121, 0.5367]$, which excludes 0.5. Ten of the sixteen individual runs do *not* resolve on their own paired interval, and two of them fall below 0.5 outright. The effect is real and the median single run fails to find it.

8 What this architecture does not do

Stated plainly, because a design document that lists only its strengths hides the more useful half.

- **No search and no lookahead at inference.** The policy is evaluated once per decision. A learned evaluator combined with search is the structural claim behind AlphaZero [3]; nothing here exercises it.
- **No opponent modelling.** All twelve auxiliary heads describe our own side. The opponent’s threat enters only through the board being visible.
- **Cross-turn history is almost absent.** Intra-turn memory exists as ordinal-indexed tokens, but cross-turn history is a handful of near-empty tokens carrying a count and a type identifier. Conditioning on action history is deliberately *not* added: it is refuted for imitation learning by [4, 5, 6] and by this project’s own removal of it with evidence.
- **No positional encoding over tokens**, which would be a category error on a set.
- **Imitation is bounded by its teacher.** Only search or self-play can exceed that bound, and neither is in the deployed path.

9 Claims withdrawn

Auxiliary heads are not worth +45 ladder points. That figure compared a multitask run against a pure-imitation run from a different pipeline. An audit of every run in this project found that *no* run has ever supplied an auxiliary target table; six auxiliary heads are constructed from a hardcoded fallback and have never received a gradient from a label. The comparison does not transfer and the claim is withdrawn. [MEASURED]

The rejection of FiLM in the ground-up design document does not follow from its cited evidence. That document rejects family-conditioned gain in favour of per-field direction vectors, on the grounds that the two readings of a shared column are “unrelated quantities needing different directions”. The measurement offered (0.0241 against 0.0003) shows one family not using the column, which is interference and is exactly what a gain repairs. The rejection may still be right, but not on that evidence, and the arm that shipped FiLM is currently ahead. [MEASURED]

A between-seed spread of 2.52 pp was mislabelled, not unsourced — and it reproduces exactly. The figure is the between-seed standard deviation of *arena win rate*, not of offline top-1, and it was handed on without naming its quantity. Recomputed from the raw per-seed artefacts, the first twelve seeds give mean 0.5253, SD 0.0252 (2.52 pp), standard error 0.0073, 95% interval [0.5110, 0.5396] — a bit-exact reproduction. Four further seeds have since landed; at sixteen the estimate is mean 0.5244, SD 0.0251, interval [0.5121, 0.5367], still excluding 0.5 and slightly tighter. The two bands in §7 are both correct and neither substitutes for the other; the defect was the missing label, which is the same failure as an unlabelled hash. [MEASURED]

The justification given for bootstrapped value targets does not apply. The stated reason to prefer temporal-difference targets over regression is that recorded labels are “22% noisy on pair-order replication.” That number traces to an eight-sample repeatability audit of seventeen contexts in an unrelated ranking experiment from an earlier phase of the project — not to the recorded terminal labels of the replay corpus. Good reasons to prefer bootstrapping exist (variance and credit assignment across roughly a hundred decisions per game); this is not one of them, and it should be struck. [MEASURED]

10 The corpus has no terminal state

A finding that constrains any value-function work built on recorded play. Sampling replays evenly across all forty-five days of the high-Elo corpus: [MEASURED]

replays containing a match-result log entry	0/288 = 0.0%
last observation with the game marked finished	0/360 = 0.0%
<i>any</i> observation with the game marked finished	0/144 = 0.0%
last observation with exactly one seat at zero prizes	0/360 = 0.0%
top-level win/loss reward well-formed	288/288 = 100%

The engine emits a terminal reason during live play; the replay serialisation does not preserve it, because the last recorded observation is the one the agent was asked to act on. The *outcome* is available and exact; the terminal *state* is not. Consequently a prize-margin target cannot be read from this corpus. The tempting substitute — treating the last observed prize counts as terminal — mispredicts the winner 21.4% of the time while passing every schema, range and completeness check that would normally be applied to it.

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